KARTIKEY GUPTA

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Portfolio: portfolio-umber-one-57.vercel.app

EDUCATION

Bharati Vidyapeeth's College of Engineering, Delhi 2023 – 2027
Bachelor of Technology in Computer Science CGPA: 8.8/10

PROFESSIONAL EXPERIENCE

AMITKK Digital Solutions June 2026 – July 2026
AI & Machine Learning Intern

- Built a Python lead-generation platform using PyQt5, FastAPI, Selenium, DuckDuckGo and MongoDB, scraping ∼80 LinkedIn posts and 100 websites daily.
- Integrated Gemini API to generate personalised outreach emails and comments with human approval.
- Automated scheduled scraping and email dispatch workflows using a scheduler.
- Designed project-based database architecture with deduplication, validation, and domain filtering.
- Resolved scheduler, duplicate-record, and API/database synchronisation issues.

PROJECTS

Cloud Kitchen ERP, CRM & Dispatch Platform
Node.js, PostgreSQL, Redis, BullMQ, Next.js, Python (FastAPI), Docker, CI/CD

- Building a multi-tenant, metadata-driven ERP where users create entities, forms and workflows in a drag-and-drop Next.js UI, with RBAC and audit logs.
- Designing inventory, order and billing modules with transactional stock reservation and a stock ledger.
- Integrating my Random Forest prep-time model (MAE 3.50 → 0.31 min) as a Python service for timed rider dispatch.
- Adding a CRM module with automated follow-ups via an event queue; tested with Jest, Supertest and pytest.

Codebase Understanding System
Node.js, LangChain.js, LangGraph.js, RAG, PostgreSQL (pgvector), Gemini API, Next.js, Docker, CI/CD

- Building an agentic RAG platform that answers questions about a GitHub repo with file- and line-level citations.
- Parsing code with tree-sitter into function-level chunks; building hybrid retrieval (vector + keyword) with reranking.
- Orchestrating a LangGraph.js agent (plan, retrieve, read, answer) with tool-calling to search code and trace calls.
- Designing webhook-triggered incremental indexing on an async job queue (BullMQ, Redis), with an evaluation harness for retrieval accuracy.

Extensible RPG Engine
JavaScript, Node.js, HTML5 Canvas, Gemini API, Docker

- Building a data-driven RPG engine with an entity system, turn-based combat, quests and save/load state.
- Designing a mod API for items, enemies and abilities, with versioning and unit tests.
- Shipping an LLM-powered NPC dialogue extension with personalities and multi-turn memory.

ACHIEVEMENTS

- Co-authored and presented "A Systematisation of Adversarial Multi-Agent Learning in Cybersecurity for Web Security" at AKGEC 2026 (3rd International Conference on Green Energy and Computing). [Paper]
- Organised BrainBot Ideathon (CodeChef, BVCOE) and CodeJam Talks (ACM-BVP).
- Solved 100+ data structures and algorithms problems on LeetCode.

TECHNICAL SKILLS

Languages	JavaScript, Python, SQL, C, C++
Backend	Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT
AI & Agents	RAG, LangChain.js, LangGraph.js, Gemini API, Tool-calling, Scikit-learn, Pandas
Frontend	Next.js, React.js, HTML, CSS
Databases	PostgreSQL, MongoDB, MySQL, Redis
DevOps & Testing	Docker, GitHub Actions (CI/CD), Cloud Deployment, Git, Jest, Supertest, pytest